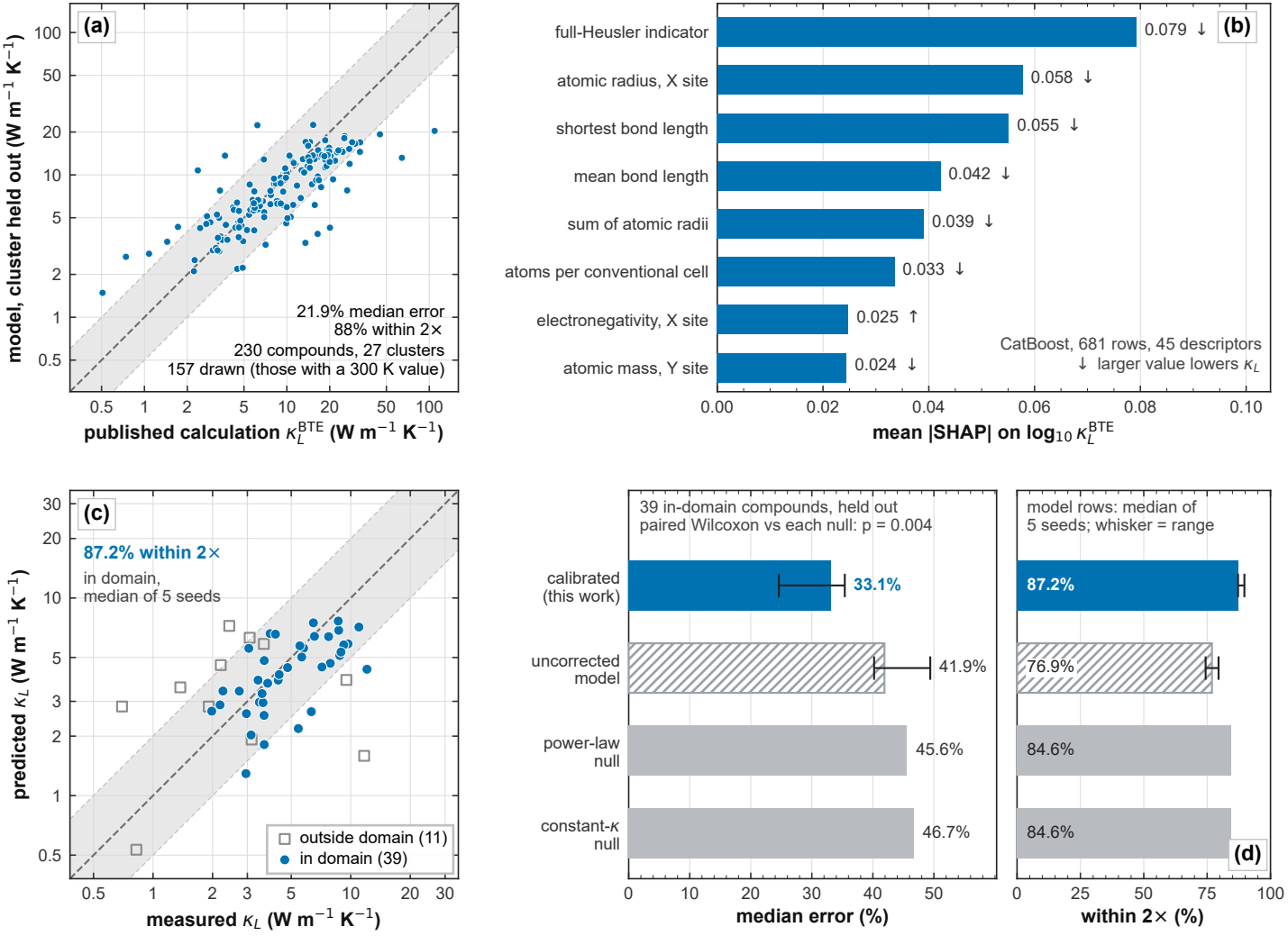

(a)
model, cluster held out (W m^{-1} K^{-1})
published calculation κ_L^{BTE} (W m^{-1} K^{-1})
21.9% median error
88% within 2×
230 compounds, 27 clusters
157 drawn (those with a 300 K value)
(b)
full-Heusler indicator
atomic radius, X site
shortest bond length
mean bond length
sum of atomic radii
atoms per conventional cell
electronegativity, X site
atomic mass, Y site
0.079 ↓
0.058 ↓
0.055 ↓
0.042 ↓
0.039 ↓
0.033 ↓
0.025 ↑
0.024 ↓
CatBoost, 681 rows, 45 descriptors
↓ larger value lowers κ_L
mean |SHAP| on log$_{10}$ κ_L^{BTE}
(c)
87.2% within 2×
in domain,
median of 5 seeds
outside domain (11)
in domain (39)
predicted κ_L (W m^{-1} K^{-1})
measured κ_L (W m^{-1} K^{-1})
39 in-domain compounds, held out
paired Wilcoxon vs each null: p = 0.004
calibrated (this work)
uncorrected model
power-law null
constant-κ null
33.1%
41.9%
45.6%
46.7%
median error (%)
model rows: median of
5 seeds; whisker = range
87.2%
76.9%
84.6%
84.6%
(d)
within 2× (%)